

Basic OS implementation questions

S.No	Topic	Simple Lab Question	Main APIs
1	File Handling	Create a file named test.txt and write the text "Hello Linux" into it.	open(), write(), close()
2	Process	Display the Process ID (PID) of the running program.	getpid()
3	Child Process	Create a child process and print a message from both the parent and child.	fork()
4	Threads	Create a thread that prints "Hello Thread".	pthread_create(), pthread_join()
5	Mutex	Create two threads and use a mutex before printing a message.	pthread_mutex_lock(), pthread_mutex_unlock()
6	Semaphore	Create two threads and use a semaphore to allow only one thread to print at a time.	sem_init(), sem_wait(), sem_post()
7	Unnamed Pipe	Send the message "Hello Child" from the parent process to the child process through a pipe.	pipe(), read(), write()
8	Named Pipe (FIFO)	Write a message into a named pipe and read it from another program.	mkfifo(), open(), read(), write()
9	Shared Memory	Store the text "Hello Shared Memory" in shared memory and display it.	shmget(), shmat()
10	Message Queue	Send the message "Hello Queue" using a message queue and receive it in another process.	msgget(), msgsnd(), msgrcv()

S.No	Topic	Simple Lab Question	Main APIs
11	TCP Socket	Create a TCP server that receives and displays a message sent by a client.	socket(), bind(), listen(), accept(), send(), recv()
12	UDP Socket	Send a message from a UDP sender and display it at the receiver.	socket(), sendto(), recvfrom()